

Week 7 Project — Unsupervised Analysis with Rigorous Evaluation

Question

Can unsupervised clustering and anomaly detection identify unusual insurance claims that should be prioritised for manual fraud review, and how reliable is that triage result?

Dataset

insurance_claims.csv

The dataset contains 8,000 insurance claims with numeric and categorical claim information and a fraud label used only for final evaluation.

Methods

Clustering

K-Means clustering was tested with K values from 2 to 6.

The final analysis used K=6 because the six-cluster solution was highly stable across five random seeds.

Anomaly Detection

Isolation Forest was used to rank unusual claims.

The fraud label was not used during fitting.

Evaluation

The analysis included:

- ⑩ Stability across five random seeds
- ⑩ Independence assessment
- ⑩ Null-case evaluation
- ⑩ Sensitivity analysis
- ⑩ 2D PCA visualisation
- ⑩ Fraud-rate baseline
- ⑩ AUC
- ⑩ Precision @ K
- ⑩ Recall @ K
- ⑩ Subgroup analysis
- ⑩ False-positive analysis

Headline Finding

The six-cluster segmentation was highly stable across random seeds.

The Isolation Forest anomaly ranking produced an AUC of approximately 0.668 on the full dataset. Approximately 9% of the top 100 unusual claims were fraudulent.

This means the anomaly ranking identifies unusual claims, but unusual claims are not the same thing as fraudulent claims.

How to Run

Place `insurance_claims.csv` in the same directory as the notebook.

Open the notebook and run all cells from top to bottom using a fresh Python kernel.

Required packages:

⑩ pandas

⑩ numpy

⑩ matplotlib

⑩ scikit-learn

Main Limitation

The dataset does not contain a customer identifier, so it is not possible to verify whether multiple claims belong to the same customer.

The anomaly detector can also miss fraudulent claims that look normal according to the available features.

Main Decision

The anomaly ranking should be used as a manual review queue rather than an automatic fraud decision.